

REQUIREMENTS DOCUMENT

Multi-Tenant Organizational Wiki Platform

Centralized document management with role-based and group-based access control

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1 Overview & Purpose

The organization needs a centralized, searchable knowledge base ("the Wiki") where teams can author, store, and manage documents. The platform must support multiple separate organizations (tenants), each with fully isolated users, groups, and content. Within an organization, access to content is controlled by role (what a user is allowed to do) and by group membership (what content a user can see by default).

This document defines functional and non-functional requirements for the initial release (v1) of the platform. It is intended for engineering, design, and product stakeholders to align on scope before implementation planning begins.

2 Goals & Success Metrics

Goals

- Provide a single source of truth for organizational documentation, replacing scattered files across drives, chat threads, and email.
- Ensure strict data isolation between separate organizations (tenants) sharing the same platform.
- Make content easy to find via full-text and metadata search.
- Allow controlled authoring with a proper rich-text editor, not just file uploads.
- Support native handling of PDF documents — upload, storage, and in-browser viewing without download.
- Enforce access control by role and by group so sensitive content is only visible to the right people by default.

Success Metrics (indicative — to be finalized with stakeholders)

Metric	Target (v1)
Search result relevance (top-3 click rate)	≥ 70%
Time to publish a new document	< 5 minutes for a standard doc
PDF preview load time (P95)	< 3 seconds for files ≤ 25MB
Cross-tenant data leakage incidents	0 (hard requirement)
Admin time to onboard a new team/group	< 10 minutes

3 Definitions & Key Concepts

Term	Definition
Organization (Tenant)	A top-level, fully isolated customer/company account. Users, groups, documents, and settings belong to exactly one organization.
User	An individual account within an organization, assigned one role and one or more groups.
Role	A platform-level permission set: Admin, Team Member, or Viewer. Determines what

Term	Definition
	actions a user can perform.
Group	A collection of users (e.g. "Finance", "Engineering", "Leadership") used to control default visibility of documents and spaces.
Space / Workspace	A top-level container for related documents (e.g. "HR", "Product", "Engineering Wiki"), used to organize content and apply group-based access at a coarse level.
Document (Page)	A single wiki article authored in the rich-text editor, or an uploaded PDF file, that lives inside a Space.
ACL (Access Control List)	The effective set of groups/roles/users permitted to view or edit a given Space or Document.

4 Multi-Tenancy Model

The platform uses a separate-organizations model: each organization is a fully isolated tenant with its own users, groups, spaces, documents, and admins. There is no cross-organization visibility of any kind by default.

4.1 Requirements

1. Each organization has a unique identifier and a dedicated namespace (e.g. subdomain such as `acme.wiki.example.com`, or an org-scoped path).
2. A user account belongs to exactly one organization in v1. (Multi-org membership per user is out of scope — see Section 12.)
3. No document, search index entry, group, or user record is ever visible or queryable across organization boundaries, including for platform-level staff without explicit break-glass tooling.
4. Each organization has at least one Admin at creation time.
5. Organization creation/provisioning is performed by a platform-level super-admin (internal Anthropic-org-equivalent staff), not self-service, unless self-service signup is explicitly requested later.
6. Branding (logo, name, primary color) is configurable per organization.

4.2 Data Isolation Approach (for engineering evaluation)

Two common approaches should be evaluated during technical design:

- Shared database, `tenant_id` column on every table + row-level security enforced at the query layer.
- Database-per-tenant or schema-per-tenant for stronger physical isolation.

Recommendation for v1: shared database with enforced row-level security and mandatory `tenant_id` scoping in every query, revisited if compliance requirements demand physical isolation for specific customers.

5 Roles & Permissions

Three roles are required for v1, scoped per organization (a user's role applies only within their own organization):

5.1 Admin

- Full control over the organization's settings and branding.
- Create, rename, and delete Spaces.
- Create, edit, and delete Groups; assign/remove users to/from Groups.
- Invite, deactivate, and change the role of any user in the organization.
- Set and override access permissions on any Space or Document, including content they don't own.
- Create, edit, delete, and restore (from trash) any document.
- View audit logs for the organization.

5.2 Team Member

- Create new documents in Spaces they have edit access to.
- Edit and delete documents they authored, or that their group has edit rights on.
- Upload PDFs to Spaces they have edit access to.
- Comment on documents (if commenting is enabled — see Section 12).
- View documents according to their group memberships and any explicit shares.
- Cannot manage users, groups, org settings, or permissions.

5.3 Viewer

- Read-only access to documents according to their group memberships and any explicit shares.
- Can search and view PDFs/pages, and download PDFs if download is permitted.
- Cannot create, edit, or delete any document, Space, group, or user.

5.4 Role Assignment Rules

1. Every user has exactly one role at a time within their organization.
2. Only Admins can change a user's role.
3. An organization must always retain at least one active Admin (block the action that would remove the last one).

6 Group-Based Access Control

Group membership determines default visibility, independent of role. Role determines what a user can do; group membership determines what content a user can see and act on by default.

6.1 Requirements

1. Admins create Groups (e.g. Finance, Engineering, All-Company, Leadership) and assign users to one or more groups.
2. Every Space has an associated set of Groups granted access (view and/or edit), configured at creation and editable later by an Admin or Space owner.

3. Default-deny model: a user can only see a Space/Document if their group membership (or an explicit individual share) grants it, or if they are an Admin.
4. A special built-in group (e.g. "All Members") can be used for organization-wide default content.
5. Documents inherit their parent Space's group permissions by default, with the option for finer-grained per-document overrides (e.g. restrict one sensitive page within an otherwise open Space).
6. Explicit individual shares (sharing a specific document with one named user regardless of their group) are supported as an override on top of group-based defaults.
7. Removing a user from a group immediately revokes the access that membership granted (subject to caching/propagation SLA — see Section 8).

6.2 Example

Space	Groups Granted Access	Access Level
Company Handbook	All Members	View
Company Handbook	HR, Leadership	Edit
Finance Reports	Finance, Leadership	View + Edit
Engineering Wiki	Engineering	Edit
Engineering Wiki	All Members	View

7 Functional Requirements

7.1 Document Editor

A rich-text ('proper') editor for authoring native wiki pages.

ID	Requirement	Priority
ED-1	Rich-text editing: headings, bold/italic/underline/strikethrough, ordered & unordered lists, blockquotes, code blocks, tables, links, images.	Must
ED-2	Auto-save while editing, with visible save status.	Must
ED-3	Version history: view and restore prior versions of a document.	Must
ED-4	Embed/attach files (including PDFs) inline within a page.	Must
ED-5	Page metadata: title, tags/labels, owner, last-edited-by, last-edited-at.	Must
ED-6	Nested/child pages within a Space (hierarchical structure).	Should
ED-7	Real-time collaborative editing (multiple users simultaneously).	Could
ED-8	Comments and inline discussion threads on a page.	Should
ED-9	Markdown import/export.	Could
ED-10	Draft vs. Published states for a page.	Should

7.2 PDF Upload & Viewer

ID	Requirement	Priority
PDF-1	Upload PDF files into a Space, subject to the uploader's edit permission on that Space.	Must
PDF-2	Max file size configurable per organization (default suggestion: 100MB per file).	Must
PDF-3	In-browser PDF viewer — no download required to read the document.	Must
PDF-4	Viewer supports page navigation, zoom, and search-within-document.	Must
PDF-5	Uploaded PDFs are indexed for full-text search (see 7.3).	Must
PDF-6	Admins can restrict download/print of a given PDF, leaving view-only access.	Should
PDF-7	Replace/version a PDF while preserving prior versions.	Should
PDF-8	Virus/malware scanning on upload before the file is made available.	Must
PDF-9	Thumbnail/preview generation for PDFs shown in listings and search results.	Should

7.3 Search

ID	Requirement	Priority
SR-1	Full-text search across all native wiki pages and PDF content, scoped strictly to the user's organization.	Must
SR-2	Search results respect the searching user's group-based and role-based access — no result leakage for content the user cannot view.	Must
SR-3	Filter search by: Space, author, date range, tag/label, file type (page vs. PDF).	Should
SR-4	Ranking considers title matches, recency, and (optionally) view/click popularity.	Should
SR-5	Search-as-you-type with instant suggestions.	Should
SR-6	Highlighted matching snippets in results, including snippets from within PDF text.	Should
SR-7	Typo tolerance / fuzzy matching.	Could

7.4 Organization & Navigation

ID	Requirement	Priority
NAV-1	Spaces as the top-level content container; each Space has a name, description, icon, and group-based ACL.	Must
NAV-2	Sidebar navigation showing Spaces and pages the current user has access to.	Must
NAV-3	Breadcrumbs reflecting page hierarchy.	Should
NAV-4	"Recently viewed" and "Recently updated" lists per user.	Should
NAV-5	Favorites/bookmarks for quick access to frequently used pages.	Could

ID	Requirement	Priority
NAV-6	Trash/recycle bin for deleted content, restorable by Admins within a retention window.	Must

7.5 Admin & User Management

ID	Requirement	Priority
AD-1	Admin console to invite users via email, view pending invites, and deactivate accounts.	Must
AD-2	Admin console to create/edit/delete Groups and manage membership.	Must
AD-3	Admin console to assign/change user roles.	Must
AD-4	Admin console to configure Space-level and document-level group access.	Must
AD-5	Audit log of key actions: permission changes, deletions, role changes, logins (retention period TBD with compliance).	Should
AD-6	SSO / SAML / SCIM support for enterprise identity provider integration.	Should
AD-7	Organization-level settings: branding, default group, file-size limits, retention policy.	Should

8 Non-Functional Requirements

8.1 Security & Compliance

- Strict tenant isolation enforced at the data layer, not just the UI (see Section 4.2).
- All data encrypted in transit (TLS) and at rest.
- Permission checks enforced server-side on every read/write, never trusted from client state.
- Uploaded files scanned for malware before storage/serving.
- Password policy / SSO support, and optional MFA for Admin accounts at minimum.

8.2 Performance

- Page load (native wiki page): P95 < 1.5s.
- PDF first-page render: P95 < 3s for files up to 25MB.
- Search query response: P95 < 500ms for indexes up to 100k documents per organization.

8.3 Availability & Reliability

- Target uptime: 99.9% (to be confirmed against business needs / SLA commitments).
- Automated backups of documents and metadata with defined RPO/RTO.

8.4 Scalability

- Support organizations ranging from ~10 users to 10,000+ users without architectural rework.
- Search and storage layers must scale horizontally per tenant load.

8.5 Accessibility & Usability

- Editor and viewer meet WCAG 2.1 AA where practical.
- Responsive layout for desktop and tablet at minimum; mobile viewing as a stretch goal.

9 Data Model (High-Level)

Illustrative entity list — not a final schema. Intended to align engineering on the core objects before detailed design.

Entity	Key Fields	Notes
Organization	id, name, subdomain, branding, created_at	Tenant root; every other entity references org_id.
User	id, org_id, email, name, role, status, created_at	role ∈ {admin, member, viewer}.
Group	id, org_id, name, description, is_default	e.g. All Members, Finance, Engineering.
GroupMembership	user_id, group_id	Many-to-many.
Space	id, org_id, name, description, icon, created_by	Top-level content container.
SpacePermission	space_id, group_id, access_level	access_level ∈ {view, edit}.
Document	id, org_id, space_id, parent_id, type, title, content_ref, owner_id, status, created_at, updated_at	type ∈ {page, pdf}.
DocumentPermission	document_id, group_id or user_id, access_level	Optional override of Space defaults.
DocumentVersion	id, document_id, version_number, content_snapshot, edited_by, edited_at	Version history.
Attachment	id, org_id, document_id, file_url, file_type, size, scan_status	Includes uploaded PDFs.
SearchIndexEntry	org_id, document_id, extracted_text, tags, updated_at	Scoped per tenant; never cross-queried.
AuditLogEntry	id, org_id, actor_id, action, target, timestamp	Permission/role/deletion events.

10 Permission Matrix

Action	Admin	Team Member	Viewer
Manage org settings	Yes	No	No
Create / delete Spaces	Yes	No	No
Create / manage Groups	Yes	No	No
Invite / deactivate users	Yes	No	No
Change user roles	Yes	No	No
Set Space/Document permissions	Yes	Own content only	No
Create documents (in accessible Space)	Yes	Yes	No
Edit documents	Any document	Own / group-editable	No
Delete documents	Any document	Own / group-editable	No
Upload PDFs	Yes	Yes (where edit access)	No
View documents	All (org-wide)	Per group / share	Per group / share
Download PDFs (if allowed)	Yes	Yes	Yes, if not restricted
Search	Org-wide	Scoped to access	Scoped to access
View audit logs	Yes	No	No

11 User Flows

11.1 Admin onboards a new team

1. Admin creates a new Group (e.g. "Marketing").
2. Admin invites users by email; invited users receive a signup/join link scoped to the organization.
3. Admin assigns each new user a role (Team Member or Viewer) and adds them to the Marketing group.
4. Admin creates a "Marketing" Space and grants the Marketing group Edit access, and "All Members" View access.

11.2 Team Member authors a document

1. User navigates to a Space they can edit and selects "New Page."
2. User writes content in the rich-text editor; content auto-saves.
3. User publishes the page, which inherits the Space's group permissions unless overridden.
4. Page becomes searchable for all users whose group/role grants access.

11.3 User uploads and shares a PDF

1. User opens a Space with edit access and selects "Upload PDF."
2. File is scanned, stored, and text-extracted for search indexing.
3. PDF appears in the Space listing with a thumbnail; other permitted users can open it in the in-browser viewer without downloading.

11.4 Viewer searches for content

1. Viewer enters a query in global search.
2. Backend executes the query scoped to the viewer's organization and filtered to Spaces/Documents their groups/role permit.
3. Results show matching pages and PDFs with highlighted snippets; the viewer opens a PDF result directly in the viewer.

12 Out of Scope (v1)

- Users belonging to multiple organizations simultaneously.
- Real-time simultaneous co-editing (single-editor-at-a-time with autosave is sufficient for v1).
- Native mobile apps (responsive web only).
- Public/anonymous (unauthenticated) sharing of documents outside the organization.
- Automated translation / multi-language content.
- E-signature workflows on PDFs.
- Granular field-level redaction within PDFs.

13 Open Questions

1. Should self-service organization signup be supported, or is provisioning always handled internally?
2. What is the required data-residency model — do any customers require region-pinned or physically isolated storage?
3. What compliance frameworks apply (SOC 2, HIPAA, GDPR) and how do they affect retention, audit-log duration, and encryption requirements?
4. Is SSO/SAML/SCIM required for launch or a fast-follow?
5. What is the expected max org size and total document volume, for capacity planning?
6. Should Viewers be able to comment, or is commenting restricted to Team Members and above?
7. What is the file-retention policy for the Trash/recycle bin before permanent deletion?
8. Do we need e-discovery / legal-hold capability for documents?

14 Assumptions

The following assumptions were made while drafting this document and should be validated with stakeholders:

- This is a web-based SaaS product (not on-premise/self-hosted) for v1.
- Hierarchy is Organization → Groups/Spaces → Documents, with role applying at the organization level.
- Document-level permissions are an override layered on top of Space-level, group-based defaults, per Section 6.
- English-only UI for v1.
- Standard modern browsers (Chrome, Edge, Firefox, Safari — current and prior major version) are the supported target; no legacy browser support required.